

ASSIGNMENT-1

MODULE-1

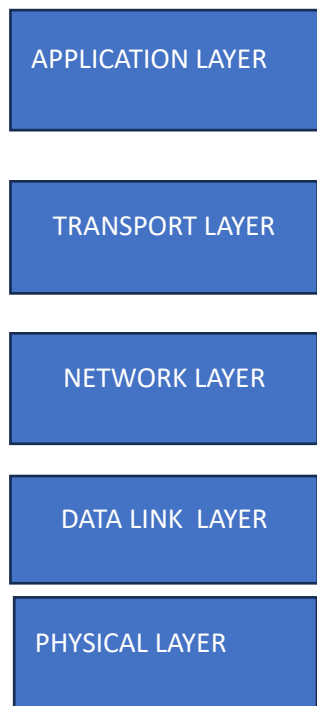
COMPUTER NETWORKS

(PCCST501)

Assignment Title: "How the Internet Delivers a Web Page"

Website chosen: Google

1. Draw the TCP/IP protocol stack and explain the role of each layer.



>1. Application Layer

The **Application Layer** provides network services directly to user applications and acts as the interface between the user and the network. It enables services such as web browsing, email, file transfer, and remote login. Common protocols include **HTTP, FTP, SMTP, and DNS**.

2. Transport Layer

The **Transport Layer** ensures reliable end-to-end communication between devices. It divides data into segments, performs error checking, controls data flow, and reassembles data at the destination. The main protocols used are **TCP** and **UDP**.

3. Network Layer

The **Network Layer** is responsible for routing data packets from the source to the destination across different networks. It provides logical addressing using **IP addresses** and selects the best path for data transmission. Common protocols include **IP** and **ICMP**.

4. Data Link Layer

The **Data Link Layer** provides reliable communication between two directly connected devices on the same network. It converts packets into frames, uses **MAC addresses** for device identification, and performs error detection and flow control. Technologies such as **Ethernet** and **Wi-Fi** operate at this layer.

5. Physical Layer

The **Physical Layer** is responsible for transmitting raw bits over the physical communication medium. It defines the electrical, mechanical, and signaling characteristics of the network, including cables, connectors, and wireless signals. Examples include **Ethernet cables, fiber optics, and radio waves**.

2. Explain the role of DNS in converting the domain name into an IP address.

>Role of DNS (Domain Name System) in Converting a Domain Name into an IP Address

The **Domain Name System (DNS)** is a naming service that translates human-readable domain names (such as www.google.com) into numerical **IP addresses** (such as **142.250.190.78**) that computers use to communicate over the Internet. Without DNS, users would have to remember IP addresses instead of website names.

Steps in DNS Name Resolution

1. The user enters a domain name (e.g., www.google.com) into a web browser.
2. The browser sends a DNS query to a **DNS resolver** to find the corresponding IP address.
3. If the resolver does not have the answer cached, it queries DNS servers (Root Server → TLD Server → Authoritative DNS Server).
4. The Authoritative DNS Server returns the IP address of the requested domain.
5. The browser uses the IP address to connect to the web server and loads the requested webpage.

3. Describe how HTTP is used to request and receive the webpage.

>HTTP (Hypertext Transfer Protocol) is the communication protocol used by web browsers and web servers to transfer web pages over the Internet. When a user enters a website address, the browser sends an HTTP request to the web server. The server processes the request and returns an HTTP response containing the requested webpage (HTML, images, CSS, etc.). The browser then displays the webpage to the user.

Steps:

1. The user enters a URL in the web browser.
2. The browser sends an HTTP request to the web server.
3. The web server processes the request and sends an HTTP response with the requested webpage.
4. The browser receives the response and displays the webpage.

User → Browser → HTTP Request → Web Server

User ← Browser ← HTTP Response ← Web Server

4. Explain how TCP provides reliable communication for HTTP.

TCP (Transmission Control Protocol) provides reliable communication for HTTP by ensuring that data is delivered accurately and in the correct order. Before transmitting data, TCP establishes a connection between the client and the server using a **three-way handshake**. It divides data into segments, assigns sequence numbers, detects errors, retransmits lost packets, and acknowledges received data. These features ensure that web pages are delivered completely and without errors.

5. Identify whether the application follows a Client–Server or Peer-to-Peer architecture and justify your answer.

A web application follows the **Client–Server architecture**. In this model, the **client** (web browser) sends requests for web pages, and the **server** processes those requests and sends the requested data back to the client. The communication is centralized because all services and resources are managed by the server, making it different from a Peer-to-Peer (P2P) network where devices communicate directly with each other.

6. Compare HTTP and FTP with at least four differences.

HTTP	FTP
HTTP stands for Hypertext Transfer Protocol .	FTP stands for File Transfer Protocol .
Used to access and transfer web pages.	Used to upload and download files between computers.
Uses port 80 (HTTP) and 443(HTTPS)	Uses port 21 for control and port 20 for data transfer
Transfers HTML pages, images, CSS, and web content.	Transfers files such as documents, images, videos, and software.
Does not require authentication for most public websites.	Usually requires a username and password for access.

7. Explain how cookies improve user experience with one real-world example.

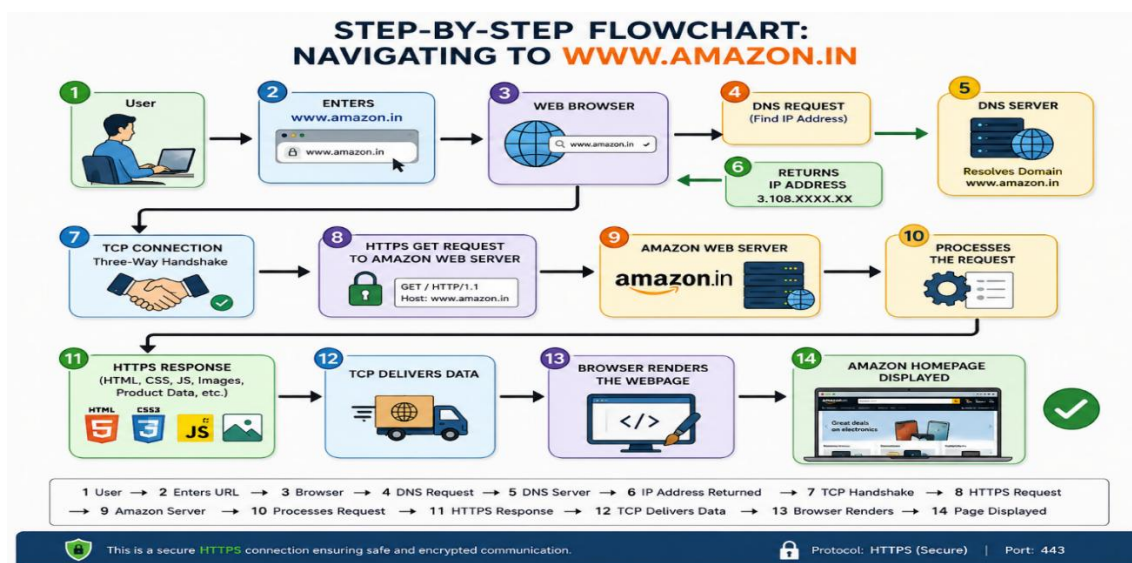
>How Cookies Improve User Experience

- **Remember Login Information:** Cookies keep users logged in, so they do not have to enter their username and password every time they visit the website.
- **Store User Preferences:** They remember settings such as language, theme (light/dark mode), and location preferences.
- **Maintain Shopping Cart:** Cookies save the items added to an online shopping cart, even if the user closes the browser and returns later.
- **Personalize Content:** Websites use cookies to display recommendations, advertisements, and content based on the user's interests and previous browsing history.

- **Real-World Example: Amazon**

- When a user visits **Amazon** and logs into their account, the website stores a cookie in the browser. This cookie remembers the user's login session, language preference, and the items added to the shopping cart. If the user closes the browser and visits Amazon again later, they usually remain logged in, and the shopping cart still contains the previously selected items. Amazon also uses cookies to recommend products based on the user's browsing and purchase history, making shopping faster and more personalized.

8. Draw a flow diagram showing the complete communication from the browser to the web server and back.



9. Write a brief conclusion (100–150 words) describing the importance of application-layer protocols in everyday Internet usage.

Application-layer protocols are essential for everyday Internet communication because they enable users to access various online services easily and efficiently. Protocols such as **HTTP/HTTPS** allow web browsing, **SMTP** and **IMAP/POP3** support email communication, **FTP** enables file transfer, and **DNS** translates domain names into addresses. These protocols define standard rules that allow different devices and applications to communicate reliably over the Internet. They also improve security, efficiency, and user experience by ensuring accurate data exchange. Without application-layer protocols, common activities such as browsing websites, sending emails, streaming videos, or downloading files would not be possible. Overall, they form the foundation of modern Internet services and make global digital communication seamless and convenient.